

COMPUTER NETWORKS

SEMESTER PROJECT REPORT — UPDATED v2.0

University Campus Network Design & Implementation

Cisco Packet Tracer Simulation

Now featuring: RIP v2 Dynamic Routing + DHCP

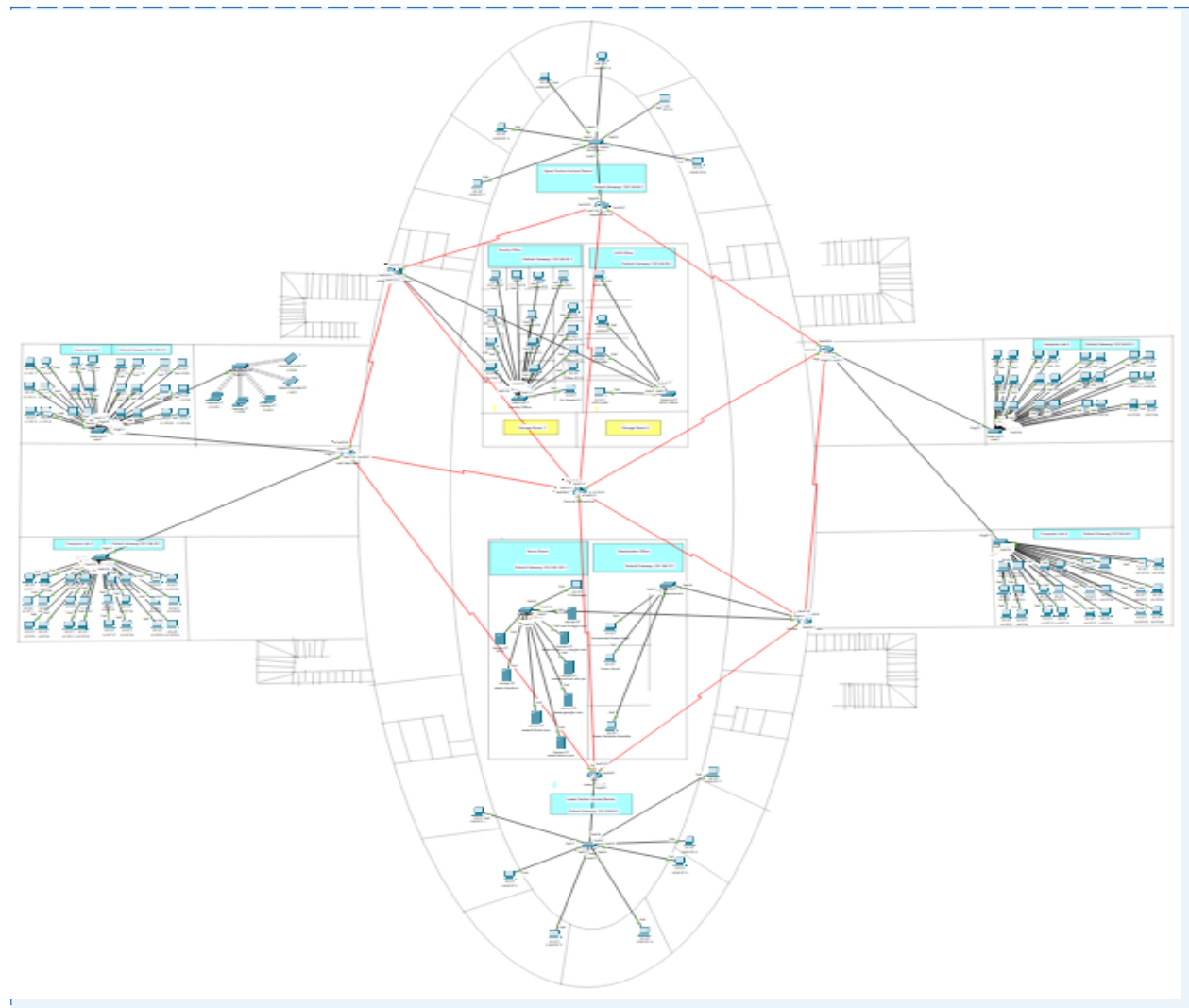
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Department	Computer Science
Course	Computer Networks (CN)
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Program	BS AI — 2nd Semester
Version	v2.0 — Updated with RIP v2 + DHCP

1. Executive Summary

✓ **UPDATED:** This is Version 2.0 of the report — updated to reflect migration from Static/Floating Static Routing to RIP v2 Dynamic Routing, and from Manual Static IP assignment to DHCP on all zone routers.

This report documents the design, configuration, and implementation of a fully functional university campus network using Cisco Packet Tracer. The network spans an entire university building and interconnects multiple zones including computer labs, lecture rooms, a faculty office, a HOD office, an exam office, a server room, and a dedicated network infrastructure backbone.

The project demonstrates practical application of core networking concepts including DHCP-based IP addressing, RIP v2 dynamic routing, fault-tolerant ring topology, DNS resolution, HTTP web hosting, email services, Access Control Lists (ACLs), and wireless connectivity.



2. Project Objectives

The primary goals of this project were:

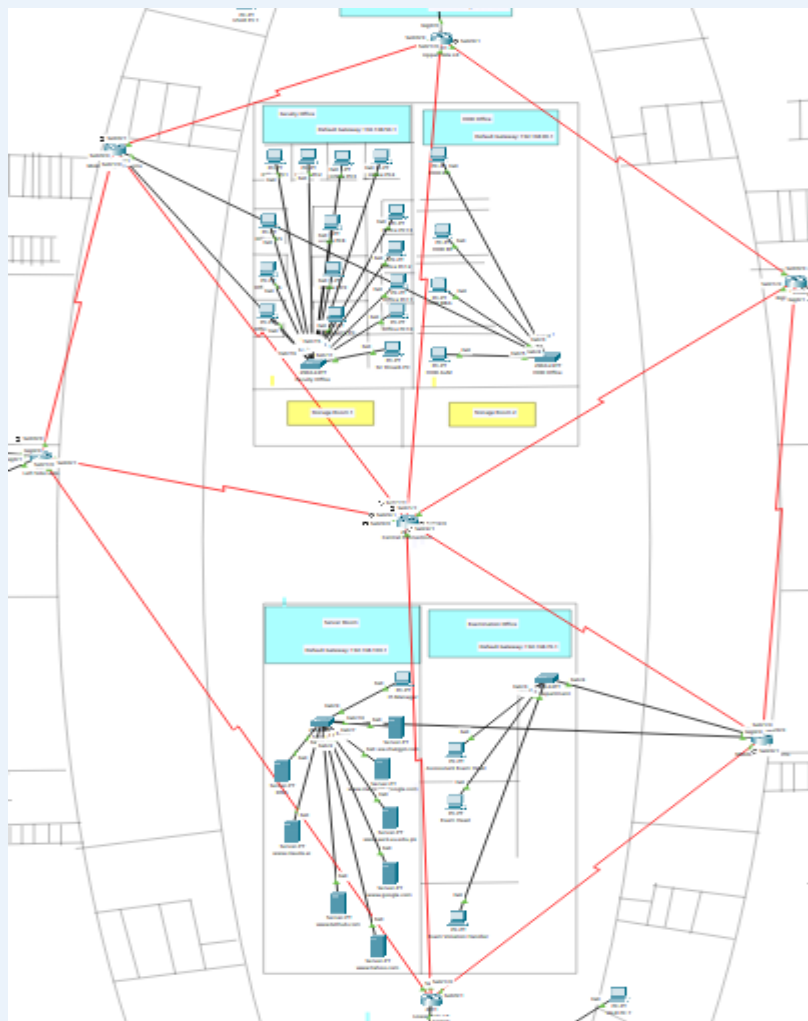
- Design a scalable, hierarchical campus network for a multi-zone university building
- Implement DHCP-based IP addressing with automatic assignment per zone
- Deploy DNS and HTTP servers to simulate real-world web hosting
- Establish secure inter-zone communication using Access Control Lists (ACLs)
- Achieve fault-tolerant routing using a ring topology with RIP v2 dynamic routing
- Configure email services for internal university communication
- Add wireless access points for mobile device connectivity

3. Network Topology Overview

The network follows a hybrid Star-Ring topology. A central router forms the backbone, while six zone routers are interconnected in a ring for redundancy. Each zone router manages its local switches and end devices.

3.1 Topology Design

- Core Layer: Central Router (2911) connected via serial links to all 6 zone routers
- Distribution Layer: Six zone routers (2911 + HWIC-2T) connected in a ring using serial DCE/DTE cables
- Access Layer: Switches (Cisco 2960) connecting end devices within each zone



3.2 Zone Router Assignments

Router	Zone Covered	LAN Subnets	Serial Ring Links
Left Side	Lab 1 + Lab 2	192.168.10.0 / 192.168.20.0	↔ Mid-Upper, ↔ Bottom
Right Side	Lab 3 + Lab 4	192.168.30.0 / 192.168.40.0	↔ Top, ↔ Mid-Lower
Mid-Upper	Faculty Office + HOD	192.168.50.0 / 192.168.90.0	↔ Left, ↔ Top
Upper Side	Top Classrooms	192.168.60.0	↔ Mid-Upper, ↔ Right
Mid-Lower	Server Room + Exam Office	192.168.100.0 / 192.168.70.0	↔ Right, ↔ Bottom
Lower Side	Bottom Classrooms	192.168.80.0	↔ Mid-Lower, ↔ Left

4. IP Addressing Scheme

✓ UPDATED: Updated in v2.0 — All end devices now use DHCP for automatic IP assignment. Only Server Room devices retain static IPs to preserve server domain mappings.

The network uses the 192.168.X.0/24 address space for LAN segments and 10.0.X.0/30 and 10.1.X.0/30 for router-to-router WAN links.

4.1 LAN Subnet Table

Zone	Subnet	Gateway	IP Assignment
Lab 1 (Left)	192.168.10.0/24	192.168.10.1	DHCP
Lab 2 (Left)	192.168.20.0/24	192.168.20.1	DHCP
Lab 3 (Right)	192.168.30.0/24	192.168.30.1	DHCP
Lab 4 (Right)	192.168.40.0/24	192.168.40.1	DHCP
Faculty Office	192.168.50.0/24	192.168.50.1	DHCP
Top Classrooms	192.168.60.0/24	192.168.60.1	DHCP
Exam Office	192.168.70.0/24	192.168.70.1	DHCP
Bottom Classrooms	192.168.80.0/24	192.168.80.1	DHCP
HOD Office	192.168.90.0/24	192.168.90.1	DHCP
Server Room	192.168.100.0/24	192.168.100.1	Static (preserved)

4.2 WAN / Router-to-Router Links

Point-to-point /30 subnets are used between all routers. Each /30 provides exactly 2 usable IPs — perfect for serial point-to-point links.

Link	Subnet	Router A IP	Router B IP
Central ↔ Left	10.1.1.0/30	10.1.1.1	10.1.1.2
Central ↔ Mid-Upper	10.1.2.0/30	10.1.2.1	10.1.2.2
Central ↔ Upper	10.1.3.0/30	10.1.3.1	10.1.3.2
Central ↔ Right	10.1.4.0/30	10.1.4.1	10.1.4.2

Central ↔ Mid-Lower	10.1.5.0/30	10.1.5.1	10.1.5.2
Central ↔ Lower	10.1.6.0/30	10.1.6.1	10.1.6.2
Ring: Left ↔ Mid-Upper	10.0.1.0/30	10.0.1.1	10.0.1.2
Ring: Mid-Upper ↔ Top	10.0.2.0/30	10.0.2.1	10.0.2.2
Ring: Top ↔ Right	10.0.3.0/30	10.0.3.1	10.0.3.2
Ring: Right ↔ Mid-Lower	10.0.4.0/30	10.0.4.1	10.0.4.2
Ring: Mid-Lower ↔ Bottom	10.0.5.0/30	10.0.5.1	10.0.5.2
Ring: Bottom ↔ Left	10.0.6.0/30	10.0.6.1	10.0.6.2

5. DHCP Configuration

✓ **UPDATED:** New in v2.0 — DHCP replaces manual static IP assignment on all end devices across all zones.

DHCP (Dynamic Host Configuration Protocol) is configured directly on each zone router. This eliminates the need for manual IP configuration on each PC — the router automatically assigns IP address, subnet mask, default gateway, and DNS server to every device that connects.

5.1 Why DHCP on Zone Routers (Not a Dedicated Server)

- Each router is directly connected to its own subnets — no ip helper-address relay needed
- Simpler configuration and easier debugging
- Each zone is independently managed — failure in one zone doesn't affect others
- Server Room kept static to preserve fixed server IPs (same concept as Google's 8.8.8.8)

5.2 DHCP Configuration — Example (Left Side Router)

```
ip dhcp excluded-address 192.168.10.1 192.168.10.10
ip dhcp excluded-address 192.168.20.1 192.168.20.10

ip dhcp pool LAB1
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.100.10

ip dhcp pool LAB2
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 192.168.100.10
```

📌 **NOTE:** IPs .1 to .10 are excluded on every pool to protect gateway and reserved addresses. DHCP assigns from .11 onwards automatically.

5.3 DHCP Pool Summary — All Routers

Router	Pool Name	Network	Excluded Range
Left Side	LAB1	192.168.10.0/24	192.168.10.1 - .10

Left Side	LAB2	192.168.20.0/24	192.168.20.1 - .10
Right Side	LAB3	192.168.30.0/24	192.168.30.1 - .10
Right Side	LAB4	192.168.40.0/24	192.168.40.1 - .10
Mid-Upper	FACULTY	192.168.50.0/24	192.168.50.1 - .10
Mid-Upper	HOD	192.168.90.0/24	192.168.90.1 - .10
Upper Side	TOPCLASS	192.168.60.0/24	192.168.60.1 - .10
Mid-Lower	EXAM	192.168.70.0/24	192.168.70.1 - .10
Lower Side	BOTTOMCLASS	192.168.80.0/24	192.168.80.1 - .10
Mid-Lower	SERVERROOM	192.168.100.0/24	Static — not DHCP

5.4 Verification

```
show ip dhcp binding
```

```

Router>enable
Router#show ip dhcp binding

```

IP address	Client-ID/ Hardware address	Lease expiration	Type
192.168.10.12	0060.5C0A.71B1	--	Automatic
192.168.10.11	0002.4A23.E5C3	--	Automatic
192.168.10.13	00E0.B019.4B84	--	Automatic
192.168.10.14	0060.5CC1.D985	--	Automatic
192.168.10.15	0000.0C85.622B	--	Automatic
192.168.10.16	0007.ECCB.D40A	--	Automatic
192.168.10.17	00D0.BC13.1313	--	Automatic
192.168.10.18	0050.0F89.4D55	--	Automatic
192.168.10.19	0001.C756.DB74	--	Automatic
192.168.10.22	00D0.58A0.6349	--	Automatic
192.168.10.20	00E0.8FB3.8137	--	Automatic
192.168.10.23	00E0.8FBC.EC6D	--	Automatic
192.168.10.21	0001.C924.D8C9	--	Automatic
192.168.10.25	000A.4136.B173	--	Automatic
192.168.10.24	00D0.BA98.AB6D	--	Automatic
192.168.10.26	0009.7C00.E625	--	Automatic
192.168.10.29	0030.F28A.C655	--	Automatic
192.168.10.33	000B.BE56.3194	--	Automatic
192.168.10.28	00D0.58A0.72A8	--	Automatic
192.168.10.31	0000.0CA3.5180	--	Automatic
192.168.10.36	0030.F296.DEE3	--	Automatic
192.168.10.37	00E0.F989.5709	--	Automatic
192.168.10.35	00E0.8FD0.C752	--	Automatic
192.168.10.30	0004.9A2C.9800	--	Automatic
192.168.10.27	000A.4190.591E	--	Automatic
192.168.10.34	0030.F29E.772E	--	Automatic
192.168.10.32	0010.1194.2269	--	Automatic
192.168.20.13	00D0.D302.AE7B	--	Automatic
192.168.20.18	0060.2F25.E216	--	Automatic
192.168.20.20	0060.3E15.3DDB	--	Automatic
192.168.20.25	000D.BDD1.43D5	--	Automatic
192.168.20.16	0001.4392.2D03	--	Automatic
192.168.20.27	000A.416C.98CA	--	Automatic
192.168.20.21	0009.7C2E.2E54	--	Automatic
192.168.20.23	0060.2F4D.D2DB	--	Automatic
192.168.20.14	000B.BEAA.B3EE	--	Automatic
192.168.20.15	0060.47E3.9196	--	Automatic
192.168.20.17	0007.EC3C.8651	--	Automatic
192.168.20.26	000B.BEB2.10D0	--	Automatic
192.168.20.11	0001.9715.CE33	--	Automatic
192.168.20.22	0001.C95A.ED6B	--	Automatic
192.168.20.19	0060.7018.32D8	--	Automatic
192.168.20.29	00E0.8FC0.1716	--	Automatic
192.168.20.28	0060.3E0C.A021	--	Automatic
192.168.20.12	0001.C973.6DCE	--	Automatic
192.168.20.24	0002.160B.4D7D	--	Automatic
192.168.20.33	0004.9AD7.E4E6	--	Automatic
192.168.20.30	0001.425A.3DE3	--	Automatic
192.168.20.32	0060.4719.B412	--	Automatic
192.168.20.31	0090.0C8E.3364	--	Automatic
192.168.20.34	0007.EC2E.C555	--	Automatic

```

Router#
Router#

```

6. Routing Configuration — RIP v2

✓ **UPDATED:** Updated in v2.0 — All static and floating static routes removed and replaced with RIP v2 dynamic routing across all routers.

6.1 Why RIP v2 over Static Routing

Static + Floating Routes (v1.0)	RIP v2 (v2.0)
30+ manual commands per router	4-5 commands per router
Manual failover via AD values	Automatic rerouting on link failure
Topology change = manual reconfiguration	RIP adapts within 30 seconds
ARP tables reset on PT reopen	RIP hellos keep ARP tables warm
Risk of misconfiguration	Self-correcting protocol

6.2 How RIP v2 Works

RIP v2 routers broadcast their routing tables every 30 seconds to all neighbors. Each router learns all available subnets automatically and selects the best path based on hop count. When a link fails, RIP detects it within 180 seconds and reroutes traffic through alternate paths — in our ring topology this means traffic flows through the ring automatically.

6.3 RIP v2 Configuration — All Routers

Central Router

```
router rip
version 2
network 10.1.1.0
network 10.1.2.0
network 10.1.3.0
network 10.1.4.0
network 10.1.5.0
network 10.1.6.0
no auto-summary
write memory
```

Left Side Router

```
router rip
```

```
version 2
network 192.168.10.0
network 192.168.20.0
network 10.0.1.0
network 10.0.6.0
network 10.1.1.0
no auto-summary
write memory
```

Right Side Router

```
router rip
version 2
network 192.168.30.0
network 192.168.40.0
network 10.0.3.0
network 10.0.4.0
network 10.1.4.0
no auto-summary
write memory
```

Mid-Upper Router

```
router rip
version 2
network 192.168.50.0
network 192.168.90.0
network 10.0.1.0
network 10.0.2.0
network 10.1.2.0
no auto-summary
write memory
```

Upper Side Router

```
router rip
version 2
network 192.168.60.0
network 10.0.2.0
network 10.0.3.0
network 10.1.3.0
no auto-summary
write memory
```

Mid-Lower Router

```
router rip
version 2
network 192.168.70.0
network 192.168.100.0
network 10.0.4.0
network 10.0.5.0
```

```
network 10.1.5.0
no auto-summary
write memory
```

Lower Side Router

```
router rip
version 2
network 192.168.80.0
network 10.0.5.0
network 10.0.6.0
network 10.1.6.0
no auto-summary
write memory
```

6.4 Verification

```
show ip route
show ip protocols
```

 **NOTE:** In show ip route, RIP-learned routes appear with 'R' prefix. The [120/X] shows AD=120 (RIP default) and X=hop count to destination.

```

ROUTER#
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 18 subnets, 2 masks
R       10.0.1.0/30 [120/1] via 10.1.1.2, 00:00:12, Serial0/0/0
           [120/1] via 10.1.2.2, 00:00:09, Serial0/0/1
R       10.0.2.0/30 [120/1] via 10.1.3.2, 00:00:13, Serial0/1/0
           [120/1] via 10.1.2.2, 00:00:09, Serial0/0/1
R       10.0.3.0/30 [120/1] via 10.1.3.2, 00:00:13, Serial0/1/0
           [120/1] via 10.1.4.2, 00:00:12, Serial0/1/1
R       10.0.4.0/30 [120/1] via 10.1.4.2, 00:00:12, Serial0/1/1
           [120/1] via 10.1.5.2, 00:00:12, Serial0/2/0
R       10.0.5.0/30 [120/1] via 10.1.5.2, 00:00:12, Serial0/2/0
           [120/1] via 10.1.6.2, 00:00:11, Serial0/2/1
R       10.0.6.0/30 [120/1] via 10.1.1.2, 00:00:12, Serial0/0/0
           [120/1] via 10.1.6.2, 00:00:11, Serial0/2/1
C       10.1.1.0/30 is directly connected, Serial0/0/0
L       10.1.1.1/32 is directly connected, Serial0/0/0
C       10.1.2.0/30 is directly connected, Serial0/0/1
L       10.1.2.1/32 is directly connected, Serial0/0/1
C       10.1.3.0/30 is directly connected, Serial0/1/0
L       10.1.3.1/32 is directly connected, Serial0/1/0
C       10.1.4.0/30 is directly connected, Serial0/1/1
L       10.1.4.1/32 is directly connected, Serial0/1/1
C       10.1.5.0/30 is directly connected, Serial0/2/0
L       10.1.5.1/32 is directly connected, Serial0/2/0
C       10.1.6.0/30 is directly connected, Serial0/2/1
L       10.1.6.1/32 is directly connected, Serial0/2/1
R       192.168.10.0/24 [120/1] via 10.1.1.2, 00:00:12, Serial0/0/0
R       192.168.20.0/24 [120/1] via 10.1.1.2, 00:00:12, Serial0/0/0
R       192.168.30.0/24 [120/1] via 10.1.4.2, 00:00:12, Serial0/1/1
R       192.168.40.0/24 [120/1] via 10.1.4.2, 00:00:12, Serial0/1/1
R       192.168.50.0/24 [120/1] via 10.1.2.2, 00:00:09, Serial0/0/1
R       192.168.60.0/24 [120/1] via 10.1.3.2, 00:00:13, Serial0/1/0
R       192.168.70.0/24 [120/1] via 10.1.5.2, 00:00:12, Serial0/2/0
R       192.168.80.0/24 [120/1] via 10.1.6.2, 00:00:11, Serial0/2/1
R       192.168.90.0/24 [120/1] via 10.1.2.2, 00:00:09, Serial0/0/1
R       192.168.100.0/24 [120/1] via 10.1.5.2, 00:00:12, Serial0/2/0

```

Central Router

```
Router>enab
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 2 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface          Send Recv Triggered RIP Key-chain
GigabitEthernet0/0    22
GigabitEthernet0/1    22
Serial0/0/1           22
Serial0/0/0           22
Serial0/1/0           22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  10.0.0.0
  192.168.30.0
  192.168.40.0
Passive Interface(s):
Routing Information Sources:
  Gateway         Distance      Last Update
  10.0.3.1         120          00:00:26
  10.1.4.1         120          00:00:07
  10.0.4.2         120          00:00:03
Distance: (default is 120)
Router#
Router#
```

Zone Router

7. Server Room Configuration

The server room is the core services hub of the campus network. It is connected to the Mid-Lower zone router and hosts all DNS and HTTP services. Any device on the network can access the server room without restrictions.

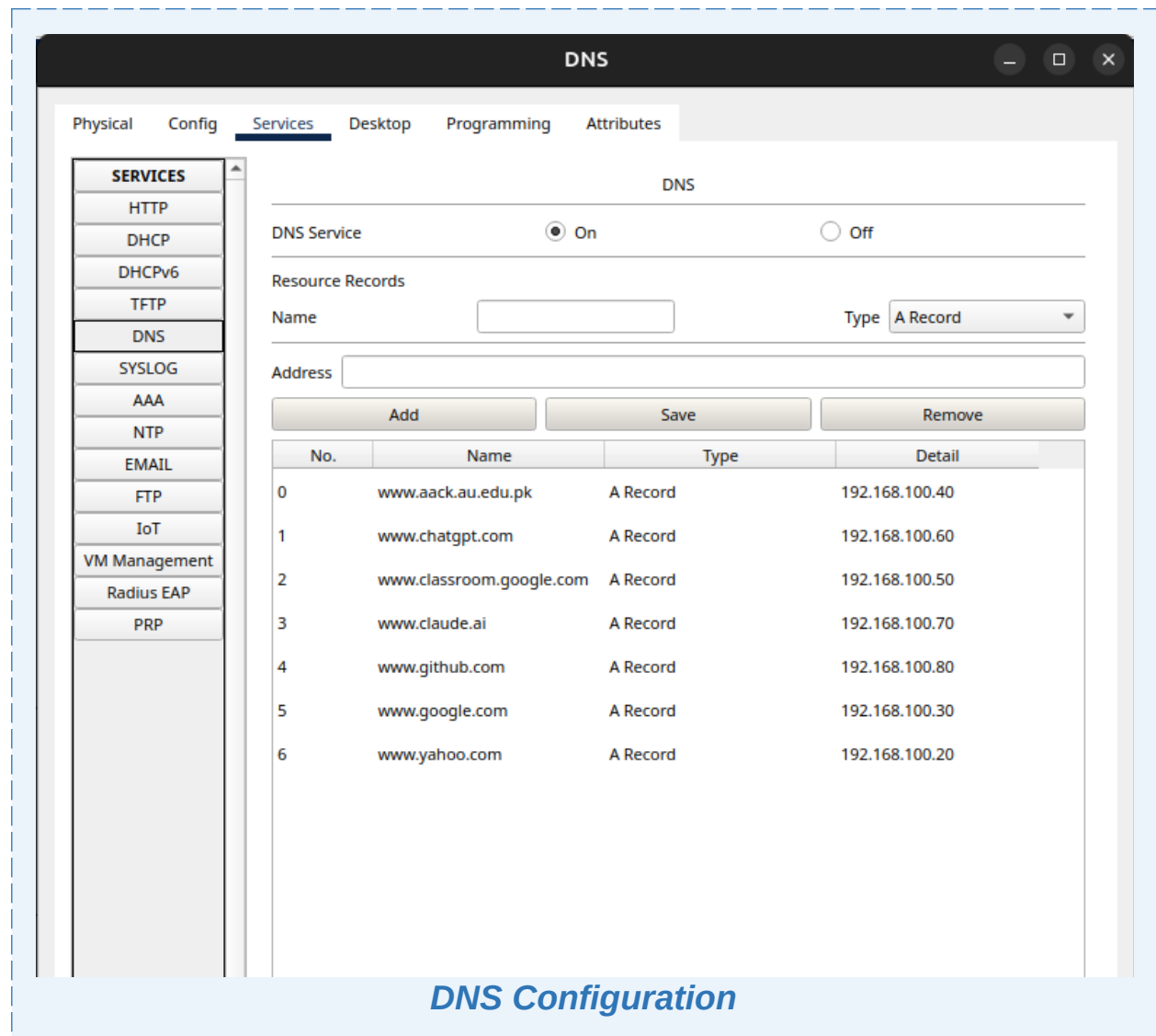
 **NOTE:** Server Room is the ONLY zone that retains static IP addressing. This ensures DNS A-records always resolve correctly — changing server IPs would break all domain name resolution across the network.

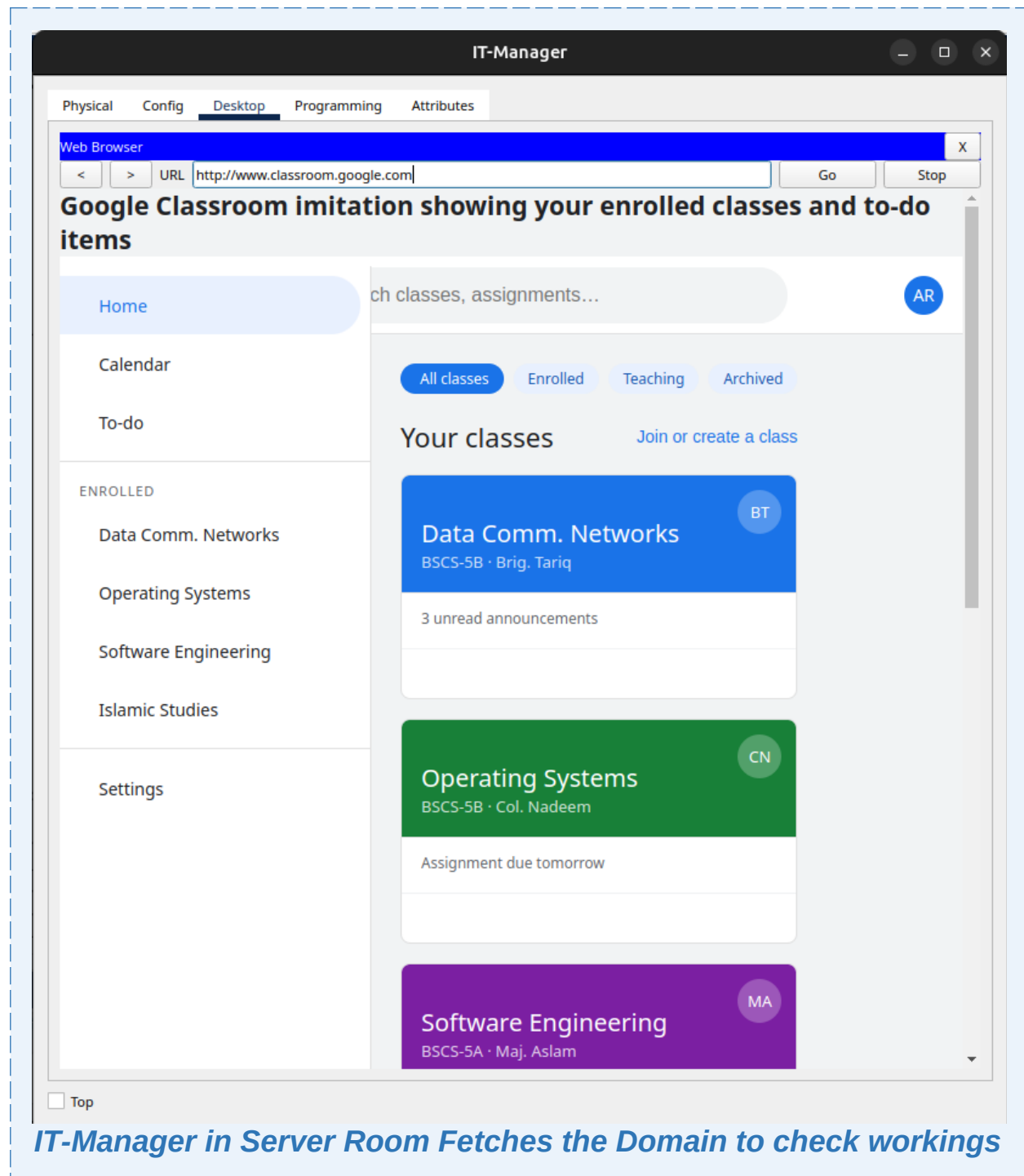
7.1 Server Inventory

Device	Domain / Role	IP Address	Service
DNS Server	DNS Resolution	192.168.100.10	DNS
Server-PT	www.chatgpt.com	192.168.100.60	HTTP
Server-PT	www.claude.ai	192.168.100.70	HTTP
Server-PT	www.yahoo.com	192.168.100.20	HTTP
Server-PT	google.com	192.168.100.30	HTTP
Server-PT	www.classroom.google.com	192.168.100.50	HTTP
Server-PT	www.aack.au.edu.pk	192.168.100.40	HTTP
Server-PT	Email Server	192.168.100.10	SMTP/POP3
PC-PT	IT Manager	192.168.100.5	Management

7.2 DNS Configuration

The DNS server holds A-records mapping each domain name to its server IP. Every end device receives the DNS server IP (192.168.100.10) automatically via DHCP dns-server option.





IT-Manager in Server Room Fetches the Domain to check workings

8. Email Service

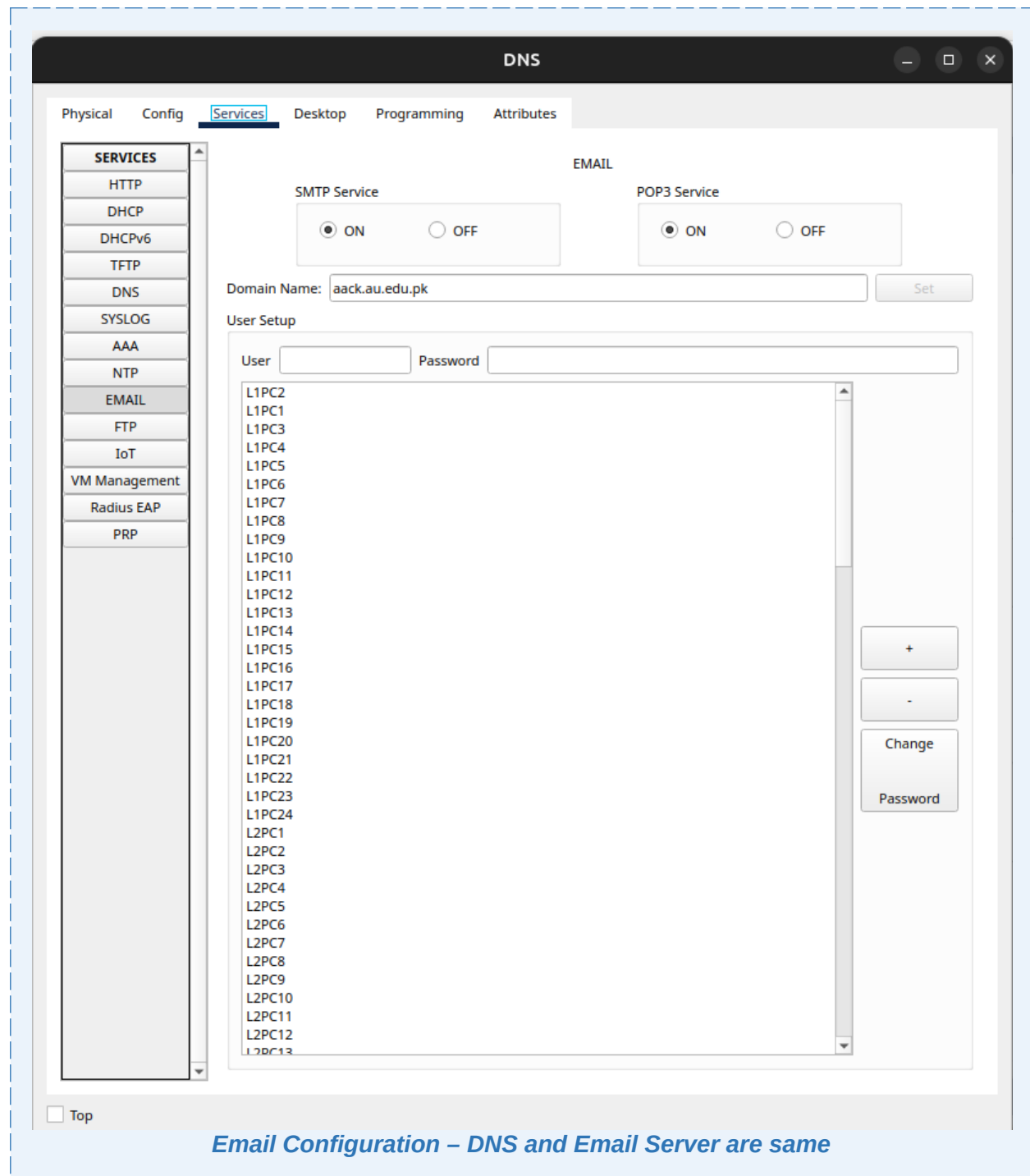
An email server is deployed in the server room providing internal communication across all university zones using SMTP for sending and POP3 for receiving.

8.1 Email Server Settings

- Domain: aack.au.edu.pk
- SMTP Port: 25
- POP3 Port: 110
- Server IP: 192.168.100.10

8.2 PC Email Client Setup

- Incoming Mail Server (POP3): 192.168.100.10
- Outgoing Mail Server (SMTP): 192.168.100.10
- Username examples: L1PC1@aack.au.edu.pk | HODCS@aack.au.edu.pk



The screenshot shows the 'Exam Head' configuration window with the 'Desktop' tab selected. The window has a title bar with standard OS controls. Below the tabs, there is a 'Configure Mail' button. The 'User Information' section contains fields for 'Your Name' (ExamHead) and 'Email Address' (ExamHead@aack.au.edu.pk). The 'Server Information' section contains fields for 'Incoming Mail Server' and 'Outgoing Mail Server', both set to 192.168.100.10. The 'Logon Information' section contains fields for 'User Name' (ExamHead) and 'Password' (masked with dots). At the bottom, there are 'Save', 'Remove', 'Clear', and 'Reset' buttons.

Exam Head

Physical Config **Desktop** Programming Attributes

Configure Mail X

User Information

Your Name: ExamHead

Email Address: ExamHead@aack.au.edu.pk

Server Information

Incoming Mail Server: 192.168.100.10

Outgoing Mail Server: 192.168.100.10

Logon Information

User Name: ExamHead

Password: ...

Save Remove Clear Reset

User Configuration

Email Received

The screenshot shows the 'HOD CS' configuration window with the 'Desktop' tab selected. The window has a title bar with standard OS controls. Below the tabs, there is a 'MAIL BROWSER' button. The 'Mails' section contains buttons for 'Compose', 'Reply', 'Receive', 'Delete', and 'Configure Mail'. Below these buttons is a table with columns 'From', 'Subject', and 'Received'. The table contains two rows of email data. The first row is from ExamHead@aack.au.edu.pk with subject 'Confirmation For the End Semester Exams!' and received on Fri Apr 24 2026 10:20:07. The second row is from HODAVM@aack.au.edu.pk with subject 'TEST!' and received on Fri Apr 24 2026 07:18:16. The 'From' field of the second row is highlighted with a blue border.

HOD CS

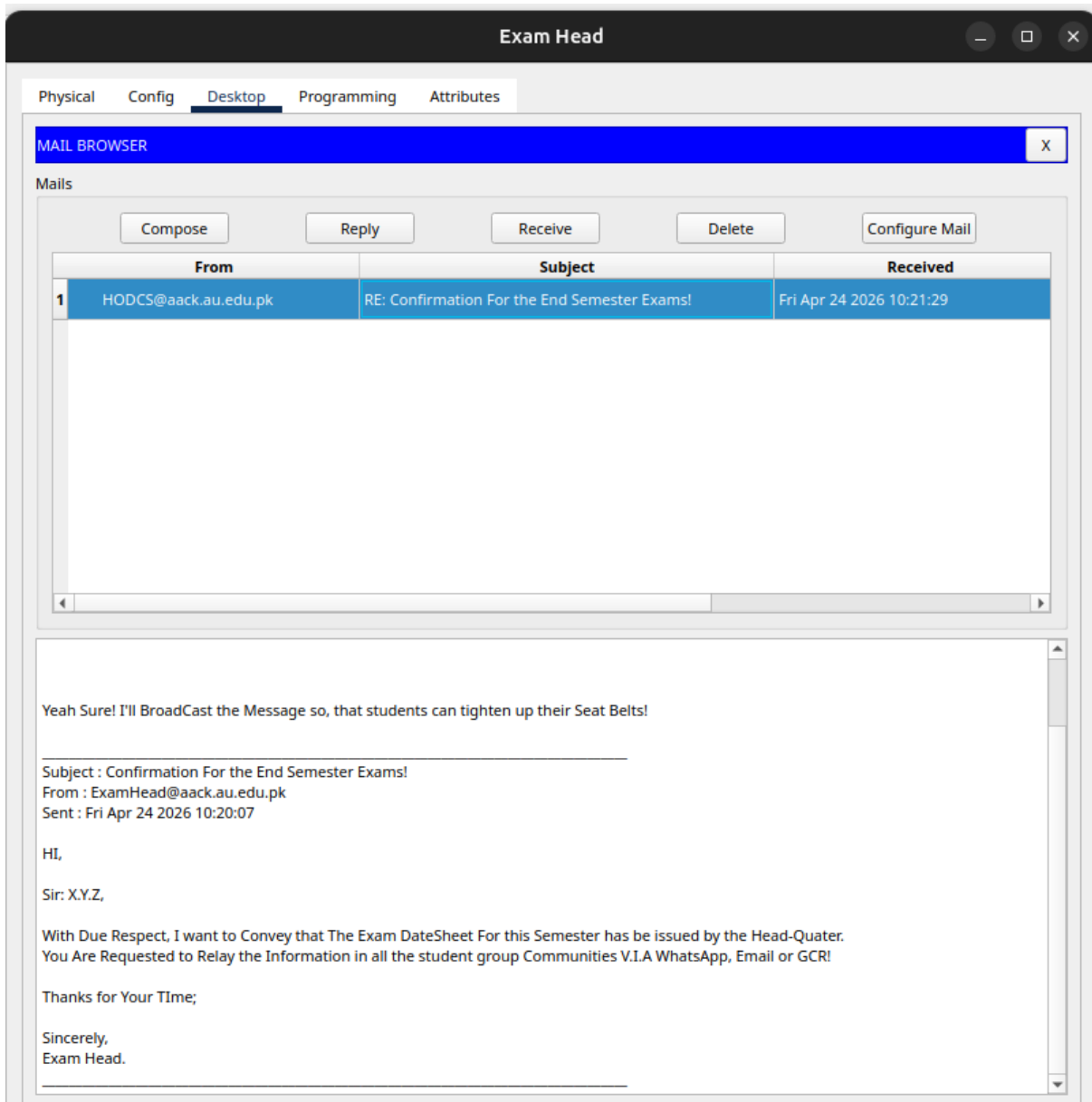
Physical Config **Desktop** Programming Attributes

MAIL BROWSER X

Mails

Compose Reply Receive Delete Configure Mail

	From	Subject	Received
1	ExamHead@aack.au.edu.pk	Confirmation For the End Semester Exams!	Fri Apr 24 2026 10:20:07
2	HODAVM@aack.au.edu.pk	TEST!	Fri Apr 24 2026 07:18:16



Received Back Email

9. Access Control Lists (ACLs)

Access Control Lists enforce a realistic university access hierarchy. The policy protects sensitive offices from unauthorized access while maintaining necessary communication paths.

9.1 Access Policy Matrix

Source \ Destination	Server Room	Faculty Office	HOD Office	Exam Office
Labs / Classrooms	✓ Allowed	✓ Allowed	✗ Blocked	✗ Blocked
Faculty Office	✓ Allowed	—	✓ Allowed	✗ Blocked
HOD Office	✓ Allowed	✓ Allowed	—	✓ Allowed
Exam Office	✗ Blocked	✗ Blocked	✓ Allowed	—
Server Room	—	✓ Allowed	✓ Allowed	✓ Allowed

 **NOTE:** HOD Office can initiate contact with Labs and Classrooms (one-way). Labs cannot reply back to HOD — models real university hierarchy where HOD broadcasts announcements but students cannot directly contact HOD.

9.2 ACL Placement

- PROTECT-HOD ACL → Applied inbound on Mid-Upper Router (GigabitEthernet0/1)
- PROTECT-EXAM ACL → Applied inbound on Mid-Lower Router (GigabitEthernet0/1)
- Left_Restrict ACL → Applied inbound on Left Router (Gig0/0 and Gig0/1)
- Right_Restrict ACL → Applied inbound on Right Router (Gig0/0 and Gig0/1)
- Top_Restrict ACL → Applied inbound on Upper Router
- Bottom_Restrict ACL → Applied inbound on Lower Router
- Faculty_Restrict ACL → Applied inbound on Mid-Upper Router (Gig0/0)
- HOD_Outbound ACL → Applied outbound on Mid-Upper Router (Gig0/1)

Middle-Upper Rooms

PhysicalConfigCLIAttributes

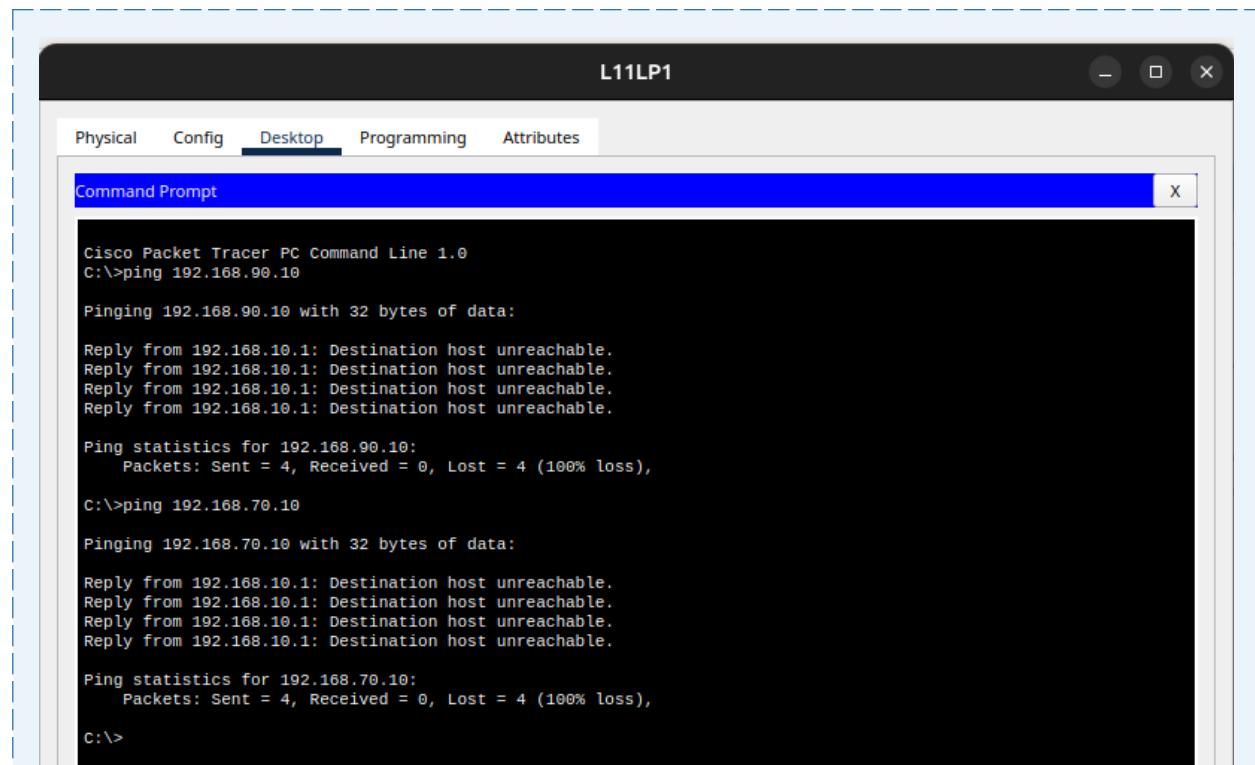
IOS Command Line Interface

```
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

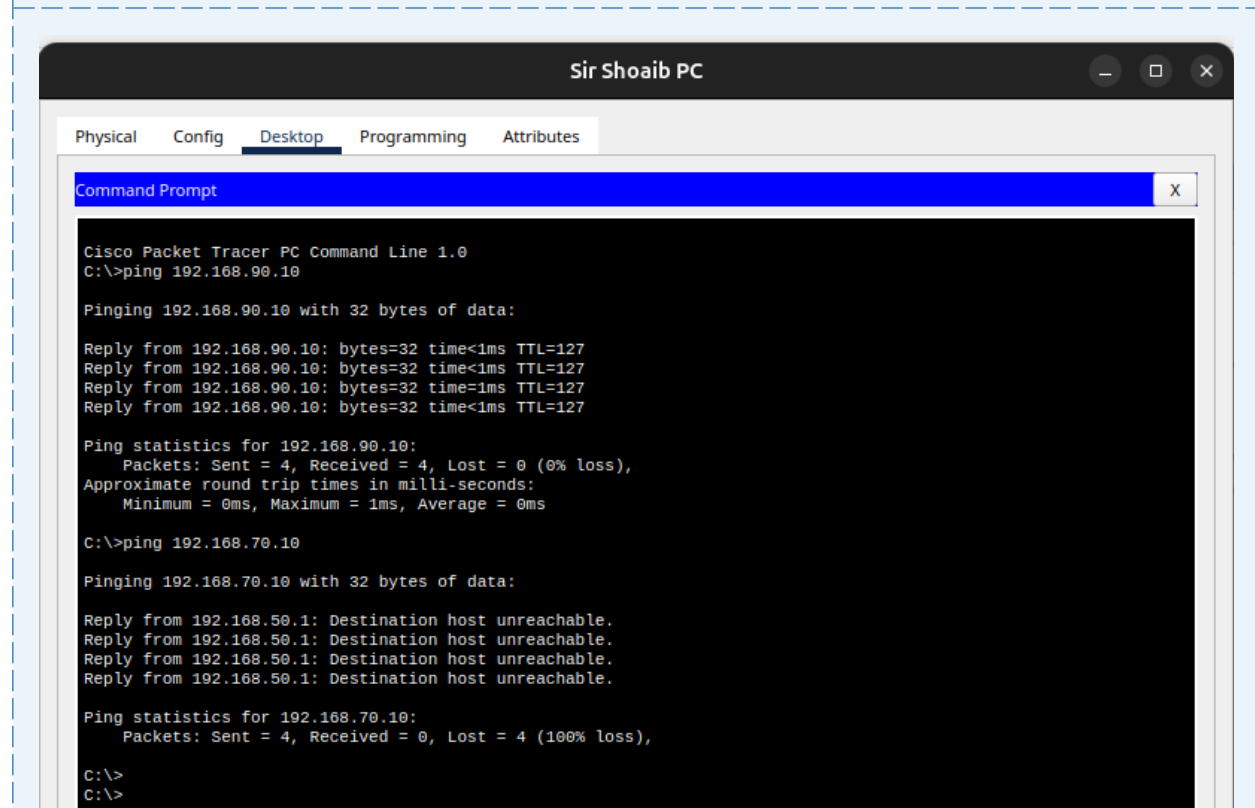
Router>
Router>enable
Router#show access
Router#show access-lists
Extended IP access list ProtectHOD
 10 deny ip 192.168.10.0 0.0.0.255 192.168.90.0 0.0.0.255
 20 deny ip 192.168.20.0 0.0.0.255 192.168.90.0 0.0.0.255
 30 deny ip 192.168.30.0 0.0.0.255 192.168.90.0 0.0.0.255
 40 deny ip 192.168.40.0 0.0.0.255 192.168.90.0 0.0.0.255
 50 deny ip 192.168.60.0 0.0.0.255 192.168.90.0 0.0.0.255
 60 deny ip 192.168.80.0 0.0.0.255 192.168.90.0 0.0.0.255
 70 permit ip any any (11 match(es))
Extended IP access list Faculty_Restrict
 10 deny ip 192.168.50.0 0.0.0.255 192.168.70.0 0.0.0.255
 20 permit ip any any
Extended IP access list HOD_Outbound
 10 permit ip 192.168.90.0 0.0.0.255 192.168.50.0 0.0.0.255
 20 permit ip 192.168.90.0 0.0.0.255 192.168.70.0 0.0.0.255
 30 permit ip 192.168.90.0 0.0.0.255 192.168.100.0 0.0.0.255
 40 permit ip 192.168.90.0 0.0.0.255 192.168.10.0 0.0.0.255
 50 permit ip 192.168.90.0 0.0.0.255 192.168.20.0 0.0.0.255
 60 permit ip 192.168.90.0 0.0.0.255 192.168.30.0 0.0.0.255
 70 permit ip 192.168.90.0 0.0.0.255 192.168.40.0 0.0.0.255
 80 permit ip 192.168.90.0 0.0.0.255 192.168.60.0 0.0.0.255
 90 permit ip 192.168.90.0 0.0.0.255 192.168.80.0 0.0.0.255
100 deny ip any 192.168.10.0 0.0.0.255
110 deny ip any 192.168.20.0 0.0.0.255
120 deny ip any 192.168.30.0 0.0.0.255
130 deny ip any 192.168.40.0 0.0.0.255
140 deny ip any 192.168.60.0 0.0.0.255
150 deny ip any 192.168.80.0 0.0.0.255
160 permit ip any any (6 match(es))

Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
Router#
```

ACL Configuration – Middle-Upper Router



Lab PC Unable to Reach HOD or Exam Office - BLOCKED



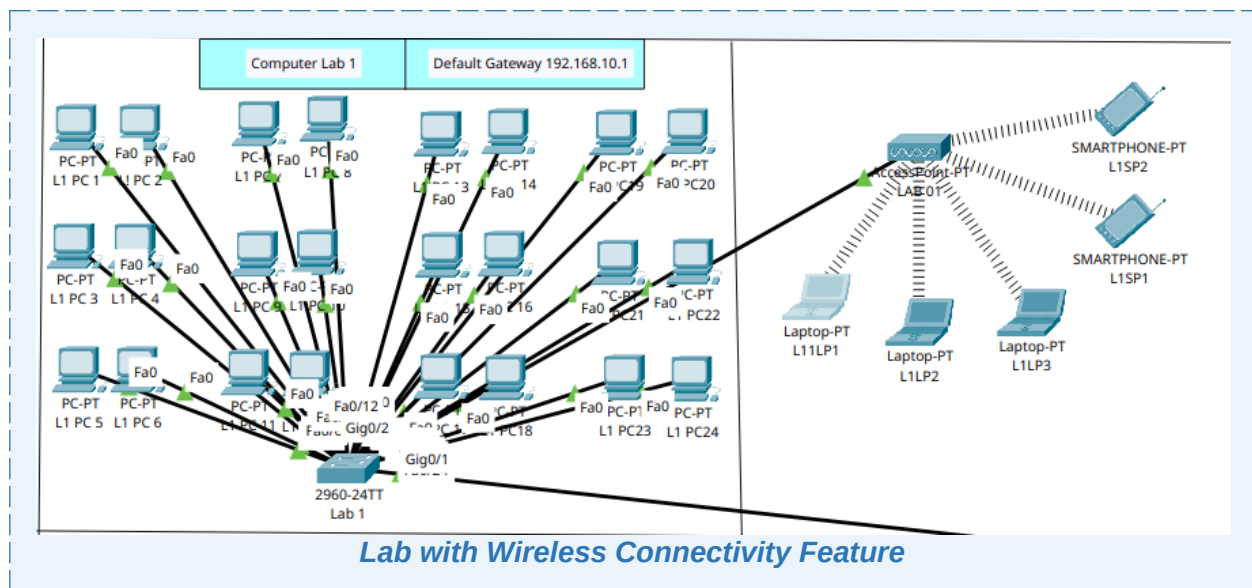
Faculty Pings to HOD (SUCCESS) and EXAM (FAIL)

10. Wireless Access Points

Wireless access points provide Wi-Fi connectivity for laptops and mobile devices. Each WAP connects to the existing zone switch sharing the same subnet and gateway as wired devices — DHCP assigns IPs to wireless devices automatically.

Location	Connected To	Subnet
Lab 01 Area	Left Zone Switch	192.168.10.0/24
Exam Office	Mid-Lower Switch	192.168.70.0/24

 **NOTE:** Only 2 WAPs added to demonstrate wireless connectivity concept. The same configuration applies to any zone where wireless is needed.



11. Testing & Verification

Test	Source	Destination	Result
Cross-zone ping	Lab 1 PC24	Faculty Office PC7	✓ PASS
Server access	Lab 1 PC24	IT Manager (100.10)	✓ PASS
DNS resolution	Any Lab PC	www.yahoo.com	✓ PASS
HTTP access	Any Lab PC	google.com	✓ PASS
DHCP assignment	Any PC (set to DHCP)	Auto IP received	✓ PASS
RIP convergence	Any router	show ip route	✓ R entries visible
ACL — Labs → HOD	Lab PC	HOD Office PC	✓ BLOCKED
ACL — Faculty → HOD	Faculty PC	HOD Office PC	✓ PASS
Failover — Central DOWN	Lab 1 PC	Faculty PC	✓ PASS via Ring

LSLR PC 5

Physical Config Desktop Programming Attributes

Command Prompt

```

C:\>
C:\>
C:\>
C:\>
C:\>ping 192.168.10.25

Pinging 192.168.10.25 with 32 bytes of data:

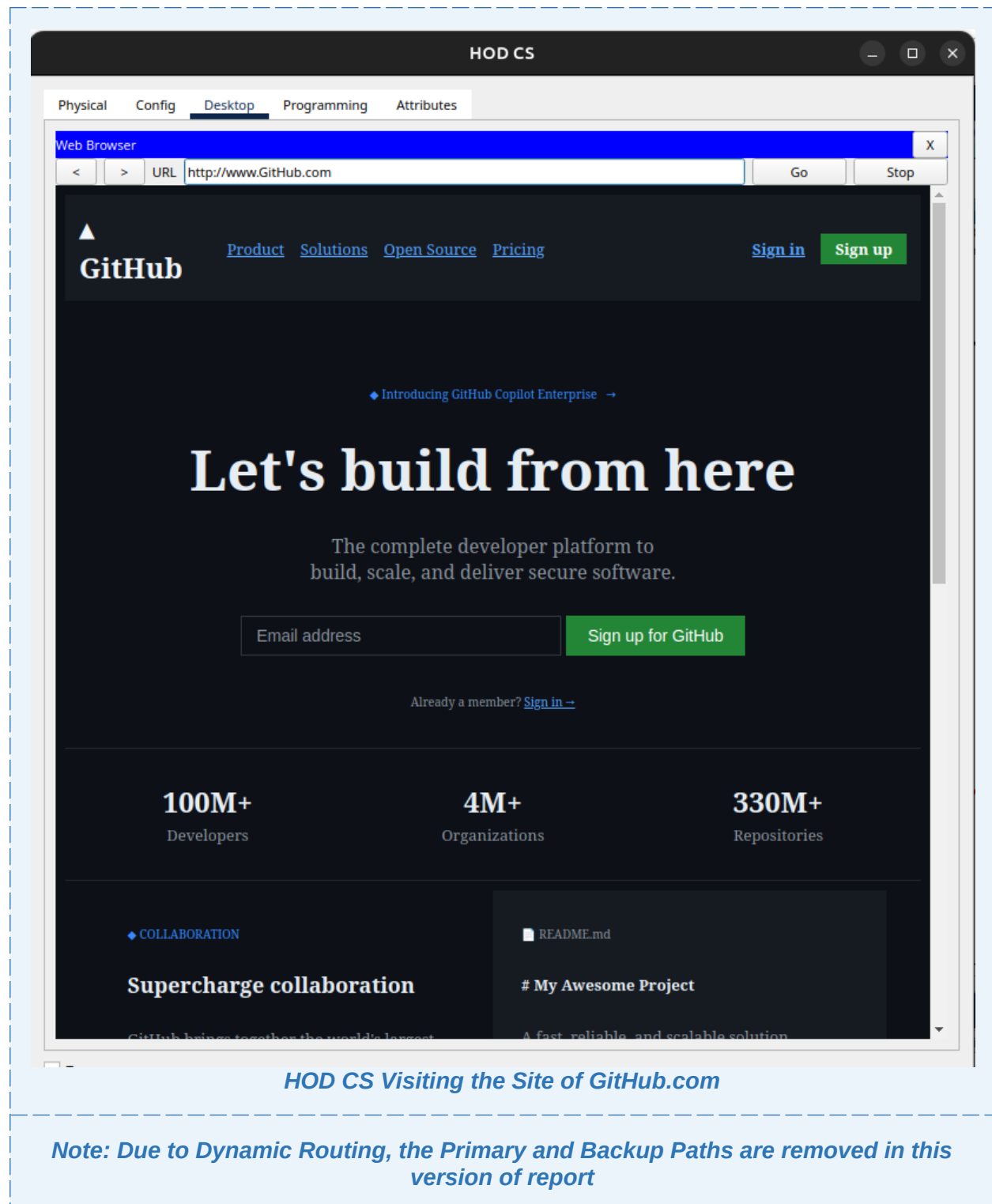
Reply from 192.168.10.25: bytes=32 time=85ms TTL=125
Reply from 192.168.10.25: bytes=32 time=2ms TTL=125
Reply from 192.168.10.25: bytes=32 time=20ms TTL=125
Reply from 192.168.10.25: bytes=32 time=3ms TTL=125

Ping statistics for 192.168.10.25:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 85ms, Average = 27ms

C:\>
C:\>
C:\>
C:\>

```

Lower-section Classroom PC pings to Lab 01 PC



12. Challenges & Solutions

Challenge Faced	Solution Applied
HWIC-1GE-SFP module not linking up	Switched to HWIC-2T serial ports for router-to-router links
IR8340 ports were Layer 2 switch ports	Replaced with standard 2911 router as central
Serial link IP overlap (wrong /8 mask)	Corrected subnet mask to 255.255.255.252 (/30)
Ring backup routes causing routing loop	Rebuilt all backup routes with correct ring order
ACL blocking HOD reply traffic from Labs	Designed as intentional one-way policy
Teammate IP addressing errors	Manually corrected all IPs and gateways zone by zone
Lower Side Router not receiving RIP updates	Central router RIP was misconfigured — fixed and all zones converged
DHCP not assigning IPs to PCs	Verified excluded-address ranges and pool network statements

13. Conclusion

This project successfully demonstrates the design and implementation of a fully functional university campus network. Version 2.0 enhances the original design by replacing manual static routing with RIP v2 dynamic routing and introducing DHCP for automatic IP management across all zones.

The hybrid Star-Ring topology with RIP v2 ensures the network remains operational even when the central router fails — RIP automatically reroutes traffic through the ring within 30 seconds, with no manual intervention required. The ACL policy models a realistic university access hierarchy.

Key technical concepts applied in this project:

- DHCP-based automatic IP addressing across 9 zones (Server Room static)
- RIP v2 dynamic routing replacing 200+ manual static route commands
- Hierarchical router design with central backbone and zone distribution
- Serial DCE/DTE WAN links with /30 point-to-point subnets
- Extended Named ACLs for granular traffic control
- DNS A-records and HTTP hosting for 7 web domains
- SMTP/POP3 email server for internal university communication
- Wireless access point deployment for mobile connectivity

 **NOTE:** Future Scope: OSPF for faster convergence than RIP, VLAN segmentation for enhanced security, QoS for traffic prioritization, and redundant server room connections.

14. Appendix — Complete Router Interface Reference

Router	Interface	IP Address	Connected To
Central	Serial0/0/0	10.1.1.1	Left Side
Central	Serial0/0/1	10.1.2.1	Mid-Upper
Central	Serial0/1/0	10.1.3.1	Upper Side
Central	Serial0/1/1	10.1.4.1	Right Side
Central	Serial0/2/0	10.1.5.1	Mid-Lower
Central	Serial0/2/1	10.1.6.1	Lower Side
Left Side	Gig0/0	192.168.10.1	Lab 1 Switch
Left Side	Gig0/1	192.168.20.1	Lab 2 Switch
Left Side	Serial0/0/0	10.0.1.1	Ring → Mid-Upper
Left Side	Serial0/0/1	10.0.6.2	Ring → Bottom
Right Side	Gig0/0	192.168.30.1	Lab 3 Switch
Right Side	Gig0/1	192.168.40.1	Lab 4 Switch
Right Side	Serial0/0/0	10.0.3.2	Ring → Top
Right Side	Serial0/0/1	10.0.4.1	Ring → Mid-Lower
Mid-Upper	Gig0/0	192.168.50.1	Faculty Switch
Mid-Upper	Gig0/1	192.168.90.1	HOD Switch
Mid-Upper	Serial0/0/0	10.0.1.2	Ring → Left
Mid-Upper	Serial0/0/1	10.0.2.1	Ring → Top
Upper Side	Gig0/0	192.168.60.1	Classroom Switch
Upper Side	Serial0/0/0	10.0.2.2	Ring → Mid-Upper
Upper Side	Serial0/0/1	10.0.3.1	Ring → Right
Mid-Lower	Gig0/0	192.168.100.1	Server Room Switch
Mid-Lower	Gig0/1	192.168.70.1	Exam Office Switch
Mid-Lower	Serial0/0/0	10.0.4.2	Ring → Right
Mid-Lower	Serial0/0/1	10.0.5.1	Ring → Bottom
Lower Side	Gig0/0	192.168.80.1	Classroom Switch
Lower Side	Serial0/0/0	10.0.5.2	Ring → Mid-Lower
Lower Side	Serial0/0/1	10.0.6.1	Ring → Left

— End of Report v2.0 —